

1	YTSS/2025/I/13								
	The cash price of a new car is \$133 000. Mr Tan trades in his old car for \$28 000 and pays a deposit for the new car. He takes up a loan which charges an overall 12% interest. He has to pay equal monthly instalments of \$800 for 7 years. Calculate the deposit that Mr Tan paid initially. [3]								
2	YTSS/2025/II/01								
	(a) The table shows the number of tourists visiting Singapore in three different years. <table><tr><td>Year</td><td>2005</td><td>2015</td><td>2023</td></tr><tr><td>Number of tourists (in thousands)</td><td>7500</td><td>11670</td><td>13170</td></tr></table> (i) Write the number of tourists in 2015 in standard form, correct to three significant figures. [1] (ii) Calculate the percentage increase in the number of tourists from 2005 to 2023. [2] (iii) The number of tourists in 2023 was 12.2% less than the number of tourists in 2024. Calculate the number of tourists in 2024. [2] (b) Water consumption rate is defined as the amount of water used per square metre in a city. In 2023, the water consumption rate of City X was 9200 litres per square metre. The total water consumption in that year was 4.6 billion litres. Calculate the area of City X in square kilometres. [2] (c) A grain of sand has an actual length of 2.4×10^{-4} m. In a magnified image, the grain of sand is shown with a length of 12 cm. (i) Find the scale used for the magnified image. Give your answer in the form $n : 1$. [1] (ii) In the same image, a small pebble is shown with a length of 18 cm. Find the actual length of the pebble. Give your answer in millimetres and standard form. [2]	Year	2005	2015	2023	Number of tourists (in thousands)	7500	11670	13170
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3	SCGS/2025/I/13								
	The value of an antique has increased by 45% from \$55 000. Tom bought it in cash with a discount of 2%. (a) Explain why the amount Tom paid for the antique is not 43% more than \$55 000. [1] (b) Calculate the actual amount Tom paid for the antique. [2]								

4	SCGS/2025/II/02																		
	The table below shows the Gross Domestic Product (GDP) of some countries and their population in 2023. <table><tr><th>Country</th><th>GDP (\$)</th><th>Population</th></tr><tr><td>Brunei</td><td>3.526×10^{10}</td><td>458 949</td></tr><tr><td>China</td><td>3.123×10^{13}</td><td>1 411 000 000</td></tr><tr><td>Luxembourg</td><td>8.853×10^{10}</td><td>666 430</td></tr><tr><td>Singapore</td><td>7.548×10^{11}</td><td>5 918 000</td></tr><tr><td>United States</td><td>2.498×10^{13}</td><td>334 900 000</td></tr></table> (a) Express the population of United States in standard form. [1] (b) The GDP of Singapore can be expressed as \$$k$ billion. Write down the value of k. [1] (c) The ratio GDP of China : GDP of Brunei can be written in the form $m : 1$. Find the value of m. Give your answer correct to the nearest integer. [2] (d) GDP per capita is defined as the GDP per person. (i) Find the GDP per capita of Luxembourg in 2023. [2] (ii) From 2022 to 2023, the GDP per capita of Luxembourg decreased by 3%. Calculate the GDP per capita of Luxembourg in 2022. Give your answer correct to the nearest thousand dollars. [2] (iii) A newspaper article claims that the GDP of Luxembourg is higher in 2022 than in 2023 due to the higher GDP per capita in 2022. Give a reason why this claim may not be true. [1]	Country	GDP (\$)	Population	Brunei	3.526×10^{10}	458 949	China	3.123×10^{13}	1 411 000 000	Luxembourg	8.853×10^{10}	666 430	Singapore	7.548×10^{11}	5 918 000	United States	2.498×10^{13}	334 900 000
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5	CCHY/2025/I/05																		
	Cindy invested \$4200 in ABN Bank for 3 years. ABN Bank pays 1.8% per year compound interest. She would have earned the same amount of interest if she had invested the same amount of money for 3 years and 6 months in HKSB Bank which pays $r\%$ per year simple interest. Calculate the value of r. [3]																		
6	CCHY/2025/I/10																		
	The base and the height of a triangle are each reduced by 40% followed by an increase by 50% Find the percentage reduction in the area of the triangle. [2]																		

7	SST/2025/I/12
	(a) Ben sold a painting to Carl at a profit of 20% and Carl then sold it to Don at a profit of 40%. If Don paid \$12800 for the painting, calculate how much Ben paid for the painting. [2] (b) Some bacteria were introduced into a culture. The number, N, of bacteria, t hours after being introduced is given by $N = 2 \times 5^t$. Find the increase in the number of bacteria after 5 hours as a percentage of the number of bacteria originally introduced. [2]
8	ACSB/2025/I/03
	Dave invests \$2000 at a rate of 1.5% quarterly compound interest. (a) Explain why the first year's interest is not 6% of \$2000. [1] (b) Calculate the amount of interest earned from the investment at the end of 3 years. [3]
9	ACSB/2025/I/05
	(a) Bob has 580 pokemon cards in his collection. He increases his pokemon cards by 115%. Find the number of pokemon cards in his collection now. [2] (b) Kate collects pokemon plushies from Japan, Thailand and Singapore. 36% of the plushies are from Japan. 25% of the remaining plushies are from Thailand. The other 24 plushies are from Singapore. Calculate the total number of plushies in her collection. [3]
10	ACSB/2025/I/11
	(a) A busy airport handled 67.7 million travelers in 2024. On average, each traveler's baggage weighed 12 kg. Find the total weight of all baggage handled in 2024, giving your answer in standard form. [2] (b) From 2024 to 2025, the number of travelers decreased by 8%. Find total number of travelers in 2025. [2]

Solution

1	Balance = \$133000 – \$28000 = \$105 000 Instalment = \$800 × 7 × 12 = \$67 200 [M1] for balance and instalment $\text{Loan} = \frac{\$67200 \times \frac{100}{112}}{112}$ = \$60000 ---[M1] Deposit = \$105 000 – \$60 000 = \$45 000 [A1]
2	(ai) $11670 \times 10^3 = 1.17 \times 10^7$ (aii) $\frac{13170 - 7500}{7500} \times 100\% = 75.6\%$ (aiii) 87.8% - 13,170,000 100% - 1,5000,000 (b) $\frac{4.6 \times 10^9}{9200} = 500000 \text{ m}^2$ = 0.5 km² (ci) $\frac{12}{2.4 \times 10^{-4} \times 100} = 500$ (cii) $\frac{18}{500} = 0.036\text{cm}$ = 0.36mm = $3.6 \times 10^{-1}\text{mm}$
3	(a) The 2% discount is based on the new value, not for \$55000. So, it cannot be just subtracted from the 45% to give 43%. (b) New value = $\frac{145}{100} \times 55000 = \\79750 Actual amount paid = $\frac{98}{100} \times 79750$ = \$78155
4	a 3.349×10^8 b $k = 754.8$ c $m = \frac{3.123 \times 10^{13}}{3.526 \times 10^{10}}$ = 885.706 ∴ $m = 886$ (nearest integer) di GDP per capita (2023) = $\frac{8.853 \times 10^{10}}{666\,430}$ = 132 842.158 ∴ GDP per capita (2023) = \$132 842.16 (2 d.p.)

	dii $\text{GDP per capita (2022)} = \frac{132\,842.158}{97} \times 100$ $= 136\,950.67$ $\therefore$ GDP per capita (2022) $= \$137\,000 \text{ (nearest thousand)}$ diii GDP per capita is also dependent on the population of Luxembourg in 2022. It is possible that the population in 2022 is low and a low GDP in 2022 may result in a higher GDP per capita in 2022 compared to 2023.
5	$\text{Interest} = 4200 \left(1 + \frac{1.8}{100} \right)^3 - 4200$ $= 230.9068944$ $\frac{4200 \times r \times 3.5}{100} = 230.9068944$ $r = 1.5707952$ $= 1.57$
6	Original area = $\frac{1}{2} \times b \times h = \frac{bh}{2}$ New area = $\frac{1}{2} \times (0.6)(1.5)b \times (0.6)(1.5)h$ $= \frac{0.81bh}{2}$ Percentage reduction = $\frac{1 - 0.81}{1} \times 100\%$ $= 19\%$
7	(a) Let the cost of the painting be \$x. $\frac{140}{100} \left(\frac{120}{100} x \right) = 12800$ $1.4 \times 1.2x = 12800$ $1.68x = 12800$ $x = \frac{12800}{1.68}$ $x = 7619.05 \text{ (2 d.p.)}$ (b) Percentage increase in the number of bacteria at the end of 5 hours $= \frac{2 \times 5^5 - 2 \times 5^0}{2 \times 5^0} \times 100\%$ $= \frac{2(5^5 - 1)}{2} \times 100\%$ $= 3124 \times 100\%$ $= 312400\%$

8	(a)	The first year's interest will be <u>more than 6% of \$2000</u> . As the interest is compounded quarterly, the <u>principal sum would increase after every 3 months for greater interest to be yielded.</u>
	(b)	$2000\left(1 + \frac{1.5}{100}\right)^{3 \times 4}$
		$= 2391.24$
		$2391.24 - 2000 = 391.24$ $= 391.24$
9	(a)	$580 \times 215\%$
		$= 1247$
	(b)	Let total number of plushies be x .
		$(1 - 0.36)(1 - 0.25)x = 24$
10		$0.48x = 24$
		$x = 50$
	(a)	$67.7 \times 10^6 \times 12$
		$= 8.124 \times 10^8$
	(b)	$\frac{92}{100} \times 67.7 \times 10^6$
		$= 6.2284 \times 10^7$ or 62284000